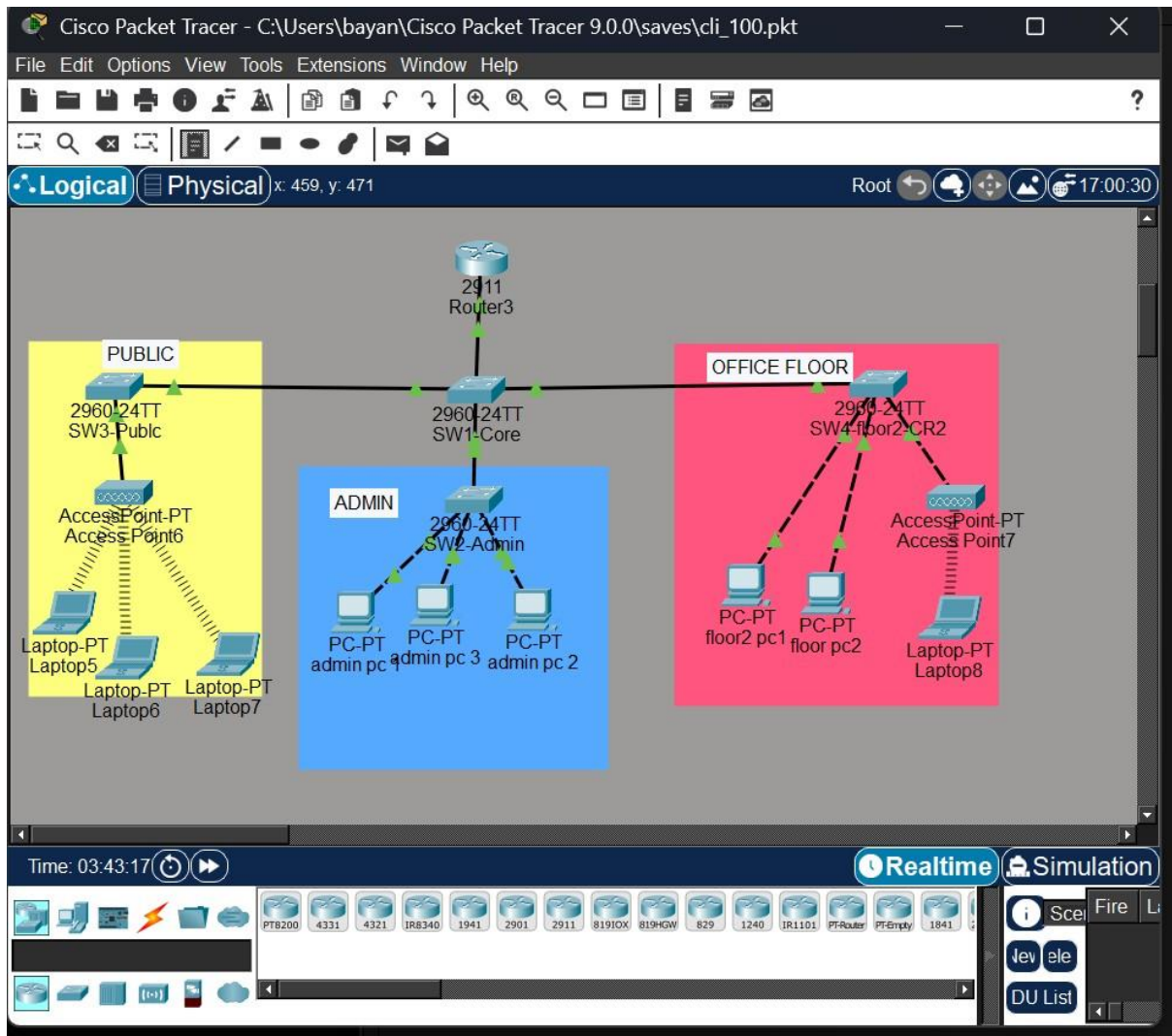


Assigned Features Implemented And Testing Evidence

Project Title: Ratlou Local Municipality Offices (Setlagole) Network Infrastructure



- **Hierarchical Architecture:** Core/Distribution layer handled by SW1-Core connected directly to access layer switches.

VLAN Database Verification: Output from `show vlan brief` on SW1-Core:

```

SW1-Core con0 is now available

Press RETURN to get started.

SW1-Core>enable
SW1-Core#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
10   Admin-Staff            active
20   Public-Area            active
30   CR2-New-Floor          active
99   Management             active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
SW1-Core#
  
```

Confirms that all operational VLANs—**VLAN 10** (Admin-Staff), **VLAN 20** (Public-Area), **VLAN 30** (CR2-New-Floor), and **VLAN 99** (Management)—are active in the local database and properly provisioned across the core infrastructure.

- **Inter-VLAN Routing:** Router-on-a-stick architecture configured on router R1Ratlou sub-interfaces using IEEE 802.1Q encapsulation.

DHCP Verification: Output from **Show ip dhcp binding** on R1-Ratlou:

```

Router3
Physical Config CLI Attributes

use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400RS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1-Ratlou>enable
R1-Ratlou#show ip dhcp binding

IP address      Client-ID/      Lease expiration    Type
Hardware address
192.168.44.7    00D0.FF91.D3C5    --                  Automatic
192.168.44.6    00E0.F77B.11C0    --                  Automatic
192.168.44.72   0030.A3ED.7513    --                  Automatic
192.168.44.73   0001.43E8.8631    --                  Automatic
192.168.44.71   000C.CF22.07AC    --                  Automatic
192.168.44.75   00E0.8FB1.5294    --                  Automatic
192.168.44.104  0006.2A00.6188    --                  Automatic
192.168.44.103  00E0.8FB1.5294    --                  Automatic
R1-Ratlou#
  
```

Displays active MAC-to-IP dynamic address leases granted by R1-Ratlou. It validates that endpoints across all subnets.

- **Trunking & Native VLAN:** IEEE 802.1Q trunking enabled across core switch links with native VLAN 99 alignment for administrative security.

Trunk Link Verification: Output from **show interfaces trunk**:

```
SW1-Core#show vlan brief
-----
VLAN Name             Status Ports
-----
1    default            active Fa0/4, Fa0/5, Fa0/6, Fa0/7
                               Fa0/8, Fa0/9, Fa0/10, Fa0/11
                               Fa0/12, Fa0/13, Fa0/14, Fa0/15
                               Fa0/16, Fa0/17, Fa0/18, Fa0/19
                               Fa0/20, Fa0/21, Fa0/22, Fa0/23
                               Fa0/24, Gig0/2
10   Admin-Staff         active
20   Public-Area         active
30   CR2-New-Floor       active
99   Management           active
1002 fddi-default         active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active

SW1-Core#show interfaces trunk
-----
Port      Mode      Encapsulation  Status  Native vlan
-----
Fa0/1     on        802.1q          trunking  99
Fa0/2     on        802.1q          trunking  99
Fa0/3     on        802.1q          trunking  99
Gig0/1    on        802.1q          trunking  99

Port      Vlans allowed on trunk
-----
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
-----
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
Gig0/1    1,10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
-----
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
Gig0/1    1,10,20,30,99

--Note--
```

Expected Result:

Demonstrates active IEEE 802.1Q trunking operational status across switch uplink interfaces. It confirms that **Native VLAN 99** is explicitly aligned across all trunk links for untagged management security while permitting VLANs 10, 20, 30, and 99.

- **Logical Segmentation:**

- **VLAN 10 (Admin-Staff):** Dedicated to internal administrative workstations and database access.
- **VLAN 20 (Public-Area):** Dedicated exclusively to public Wi-Fi access points.
- **VLAN 30 (CR2-New-Floor):** Dedicated to secondary floor wired and authenticated wireless infrastructure.

End-to-End Ping Verification: Successful ping output from Floor 2 Wireless Laptop to Gateway **192.168.44.1**:

```

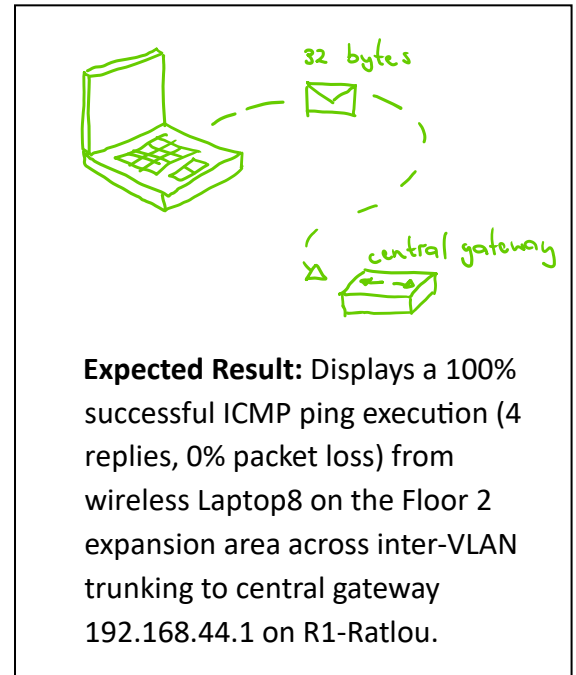
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.44.1

Pinging 192.168.44.1 with 32 bytes of data:

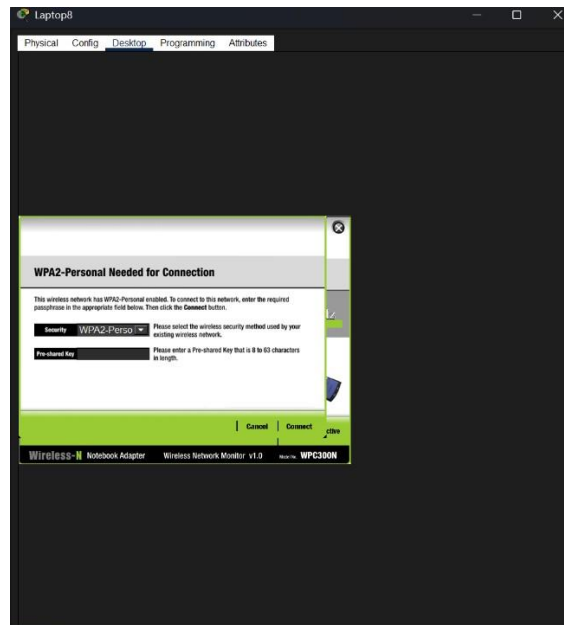
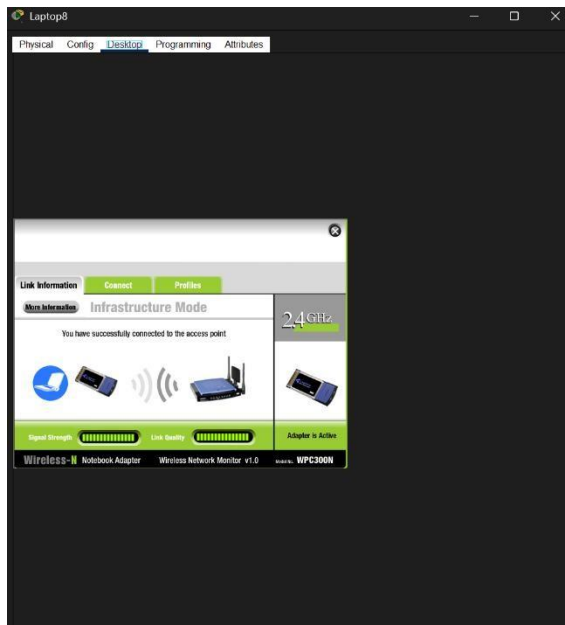
Reply from 192.168.44.1: bytes=32 time=24ms TTL=255
Reply from 192.168.44.1: bytes=32 time=26ms TTL=255
Reply from 192.168.44.1: bytes=32 time=15ms TTL=255
Reply from 192.168.44.1: bytes=32 time=20ms TTL=255

Ping statistics for 192.168.44.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 25ms, Average = 21ms
C:\>

```



- WPA2-PSK Authentication for Wireless Office floor laptop:



- **VLAN 99 (Management):** Reserved for native VLAN traffic and network infrastructure management.

Assigned Feature Implemented (Change Request 2: Floor 2 Expansion)

A. Architectural Scope & Purpose

The Floor 2 Expansion (CR2) expands the Ratlou Local Municipality network infrastructure to support a second operational office floor. The implementation deploys access switch SW4-Floor2-CR2 connected via an 802.1Q trunk link to SW1-Core, serving both wired administrative workstations (floor2 pc1, floor pc2) and wireless mobile units (Laptop8) via access point AP-Floor2.

Feature Verification Tests (CR2 Floor 2 Expansion)

DHCP Address Assignment: On floor2 pc1 & Laptop8 → Command Prompt: **ipconfig /renew**

```
floor pc2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . : FE80::2D0:BAFF:FEC3:EDD1
IPv6 Address . . . . . : ::
IPv4 Address . . . . . : 0.0.0.0
Subnet Mask . . . . . : 0.0.0.0
Default Gateway . . . . . : 0.0.0.0

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . :
IPv6 Address . . . . . :
IPv4 Address . . . . . : 0.0.0.0
Subnet Mask . . . . . : 0.0.0.0
Default Gateway . . . . . : 0.0.0.0
```

```
Laptop8
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . : FE80::2E0:8FFF:FEB1:5294
IPv6 Address . . . . . : ::
IPv4 Address . . . . . : 192.168.44.104
Subnet Mask . . . . . : 255.255.255.224
Default Gateway . . . . . : 192.168.44.97

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . :
IPv6 Address . . . . . :
IPv4 Address . . . . . : 0.0.0.0
Subnet Mask . . . . . : 0.0.0.0
Default Gateway . . . . . : 0.0.0.0
```

floor2 pc1 (FAILED / UNASSIGNED):

- **IP Address:** 0.0.0.0
- **Verdict:** floor2 pc1 failed to obtain a lease via DHCP.

Verdict: Laptop8 is working perfectly! This proves that router R1-Ratlou's DHCP pool (CR2Pool), VLAN 30 sub-interface, AP-Floor2, and switch trunking are all functional.

Fixed: changed static to DHCP

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . : FE80::2D0:BAFF:FEC3:EDD1
IPv6 Address . . . . . : ::
IPv4 Address . . . . . : 192.168.44.105
Subnet Mask . . . . . : 255.255.255.224
Default Gateway . . . . . : 192.168.44.97

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . . :
IPv6 Address . . . . . :
IPv4 Address . . . . . : 0.0.0.0
Subnet Mask . . . . . : 0.0.0.0
Default Gateway . . . . . : 0.0.0.0
```

- Router DHCP Binding
- On R1-Ratlou CLI: **show ip dhcp binding**

```
R1-Ratlou>enable
R1-Ratlou#show ip dhcp binding
IP address      Client-ID/      Lease expiration      Type
Hardware address
192.168.44.6    00E0.F77B.11C0    --                      Automatic
192.168.44.7    00D0.FF91.D3C5    --                      Automatic
192.168.44.72   0001.43E8.8631    --                      Automatic
192.168.44.71   000C.CF22.07AC    --                      Automatic
192.168.44.73   0030.A3ED.7513    --                      Automatic
192.168.44.103  0006.2A00.6188    --                      Automatic
192.168.44.104  00E0.8FB1.5294    --                      Automatic
192.168.44.105  00D0.BAC3.EDD1    --                      Automatic
R1-Ratlou#
R1-Ratlou#
```

Active IP leases listed for the 192.168.44.96/27 pool (CR2Pool).

- VLAN & Access Port Assignment
- On SW4-Floor2-CR2 CLI: **show vlan brief**

```
SW4-Floor2-CR2>enable
SW4-Floor2-CR2#show vlan brief
VLAN Name      Status Ports
-----
1    default      active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Gig0/1, Gig0/2
30    CR2-New-Floor active Fa0/1, Fa0/2, Fa0/10
99    Management    active
1002 fddi-default  active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
SW4-Floor2-CR2#
```

Ports Fa0/1, Fa0/2, and Fa0/10 are assigned and active under **VLAN 30 (CR2-New-Floor)**.

- Trunking Alignment
- On SW4-Floor2-CR2 CLI: **show interfaces trunk**

```
SW4-Floor2-CR2>enable
SW4-Floor2-CR2#show interfaces trunk
Port      Mode      Encapsulation      Status      Native vlan
Fa0/24    on        802.1q              trunking    99

Port      Vlans allowed on trunk
Fa0/24    1-1005

Port      Vlans allowed and active in management domain
Fa0/24    1,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,30,99
SW4-Floor2-CR2#
```

Port Fa0/24 is trunking with **Native VLAN 99** and allowing VLAN 30.